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Exploration of key aroma active compounds in strong flavor Baijiu during the distillation by modern instrument detection technology combined with multivariate statistical analysis methods

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ABSTRACT

Liquid-liquid microextraction (LLME) combined with gas chromatography-mass spectrometry (GC-MS) and headspace-gas chromatography-ion mobility spectrometry (HS-GC-IMS) were used to detect the variations in volatile compounds during the distillation process of strong flavor raw Baijiu. The 3D fingerprint spectrum clearly showed a variation in volatile compounds from different distillation stages, and most alcohol-soluble and low molecular weight compounds showed a downward trend with the extension of distillation time, but water-soluble, high molecular weight and high boiling compounds showed the opposite result. Then, 50 aroma compounds were sniffed and identified by gas chromatography-olfactometry-mass spectrometry (GC-O-MS) and multivariate statistical analysis including principal component analysis (PCA) and partial least squares discriminant analysis (PLS-DA) confirmed four aroma active markers related to classification. Finally, the receiver operating characteristic (ROC) curve was further used to verify that ethyl butanoate and ethyl hexanoate were the most effective difference marker to distinguish the head Baijiu samples and butanoic acid was the most effective difference marker for distinguishing the heart Baijiu samples from the tail Baijiu samples.

1. Introduction

The annual output of Baijiu, the most popular alcoholic beverage in China, exceeded 4 million kiloliters (kL) in 2020 and has created a market profit of more than 700 billion RMB (data from the China Alcoholic Drinks Association, 2021). Due to the great diversity in the production environment, brewing materials, fermentation equipment and manufacturing processes, Baijiu is generally classified into 12 categories according to its aroma characteristics (Liu and Sun, 2018). Among them, the strong flavor type Baijiu has a unique style of "mellow fragrance, refreshing and sweet taste, especially fragrant after drinking, and endless aftertaste", which is widely loved by consumers, and its output accounts for about 70% of total Baijiu production in China (Fan and Qian, 2010; Cao et al., 2021).

The flavor characteristics of strong flavor Baijiu depend on the fermentation technology and geographical environment or microenvironment of the fermentation pit. Additionally, the distillation technique is considered to be one of the most important factors in the production of Baijiu, which is closely correlated with quality and yield. Due to the effect of mass and heat transfer, distillation destroys the balance of the ethanol-aqueous solution system that originally exists in the fermented grains. The flavor constituents of the matrix are transferred into raw Baijiu by vaporization, extraction and fractionation by distillation. The process is carried out in a batch pattern, and raw Baijiu is divided into several appropriate fractions to control quality by experienced workers (Zheng et al., 2014). Therefore, revealing the flavor features of raw Baijiu at different distillation stages may contribute to a better understanding of the effects of raw Baijiu on the quality of the final product and facilitate scientific classification.

Research on the distillation stage of strong flavor Baijiu is being gradually carried out. Li et al. found that the compositions and variations of aroma substances mainly depend on their physical and chemical properties by investigating the variations of aroma substances in the distillation process of Chinese Luzhou-flavor liquor. The aroma

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Table 1
Baijiu sample information.

No.[a]	Distillate time (min)	Ethanol Content (V/V)
1	0–2	80%
2	2–4	80%
3	4–6	80%
4	6–8	76%
5	8–10	76%
6	10–12	56%
7	12–14	49%
8	14–16	31%

[a] sample number.

substances with a low boiling point, such as ethanol, acetaldehyde, ethyl formate, and ethyl acetate, were vaporized quickly and therefore had high content in the distillate obtained in the early stage of distillation. The aroma substances with high boiling points enter into distillate mainly by extraction and, hence, their variations depend on the dissolving properties (Li et al., 2012). Lang et al. investigated the distillation characteristics of volatile aroma compounds in the distillation process of Luzhou-flavor liquor and determinded the characteristic aroma compounds in the base liquor with different distillation stages (Lang, 2015). Xu et al. analyzed the distillate properties of volatile compounds and ethanol content changes of base liquor of Luzhou-flavor Baijiu during the distillation process, and found that the main volatile compounds were ethyl acetate and other esters in the early stage of distillation; the main volatile compounds were esters including butyl butanoate and isoamyl alcohol and nonanal in the middle and early stages of distillation; esters including ethyl benzoate and alcohols including hexanol in the middle and late stages; esters including ethyl phenylacetate and acids including hexanoic acid in the tail of the distillation (Xu et al., 2019). There are also many studies similar to the above (Ding et al., 2015; Jiang et al., 2021; Zheng et al., 2014; Huang et al., 2019; Zheng et al.,2013; Zhang et al., 2018). However, papers mainly focused on the changing trend of some volatile substances, while few studies have revealed the influence of aroma active compounds on the distillation stage. Recently, He et al. explored the key aroma compounds that led to the difference in raw Baijiu samples at different distillation stages from the perspective of aroma active compounds combined with multivariate statistical analysis (He et al., 2021). However, the research object was only to select typical Baijiu samples in different distillation stages, but the whole distillation process was not systematically analyzed.

Therefore, the objectives of this study were (I) to use GC-MS and GC-IMS to research the physical and chemical properties of the raw Baijiu sample and the change of volatile compounds during the distillation process; (II) to use GC-O-MS to identify the aroma active compounds in raw Baijiu samples during the distillation process; and (III) to reveal the differences in the content of aroma active compounds at different distillation stages through multivariate statistical analysis and further explore the key aroma active compounds that cause the differences in raw Baijiu samples at different distillation stages and verify its diagnostic ability to find the most effective difference markers. We hope this research could provide a theoretical basis for subsequent monitoring and improve the quality of the base Baijiu.

2. Material and methods

2.1. Samples

Sampling was obtained from Bandaojing Baijiu Co., Ltd. (Shandong, China). A total of 8 fresh raw Baijius were collected from different distillation times of a layer of fermented grains taken every 2 min (50 mL sampled at each time). Samples were sealed at 4 °C until analysis. The samples are detailed in Table 1.

2.2. Chemicals

All standards were of GC quality with at least 95% purity. Compounds, including ethyl butanoate, butyl acetate, 2-methylpropanol, ethyl pentanoate, 1-butanol, ethyl 4-methylpentanoate, 3-methyl-1-butanol, ethyl hexanoate, hexyl acetate, furfuryl ethyl ether, cyclopentanol, propyl hexanoate, ethyl lactate, 1-hexanol, ethyl 2-hydroxybutanoate, hexyl butanoate, propyl lactate, ethyl octanoate, acetic acid, isopentyl hexanoate, furfural, 2-acetylfuran, furfuryl acetate, ethyl nonanoate, ethyl DL-2-hydroxy-4-methylpentanoate, hexyl hexanoate, butanoic acid, ethyl decanoate, phenylacetaldehyde, 2-furanmethanol, 3-methylbutanoic acid, diethyl cutaneious, 2-methoxyhexyl acetate, 3-methylthiopropanol, pentanoic acid, hexanoic anhydride, diethyl glutarate, ethyl phenylacetate, 4-methylpentanoic acid, phenethyl acetate, hexanoic acid, ethyl 3-phenylpropanoate, 2-phenylethanol, heptanoic acid, 2-methoxy-*p*-cresol, phenol, 4-ethyl-2-methoxyphenol, *p*-cresol, methyl palmitate and ethyl linolenate, were purchased from Sigma-Aldrich (Shanghai, China). Pentyl acetate, diethylacetic acid, and 4-octanol, from Sigma-Aldrich (Shanghai, China), were used as internal standards and labeled as IS1, IS2, and IS3, respectively. A C_7 −C_{40} *n*-alkane mixture (Sigma-Aldrich, Shanghai, China) was employed for determining the linear retention index (LRI). Absolute ethanol, sodium chloride and dichloromethane were obtained from China National Pharmaceutical Group Corp. (Shanghai, China).

2.3. Aroma compound liquid-liquid microextraction (LLME) and fractionation

LLME was performed to extract aroma active compounds from Baijiu. About 2.0–4.0 mL Baijiu samples were diluted to 15% in ethanol (v/v) with ultrapure water (Millipore, Bedford, MA), and NaCl was added to saturation. Internal standards (pentyl acetate, diethylacetic acid, and 4-octanol) were also added according to the proportion of 1 mL of original Baijiu with 20 ppm internal standards. Then, 5.0 mL of the above sample was transferred to a 10.0 mL centrifuge tube, and 1.0 mL of dichloromethane (extraction solvent) was injected into the Baijiu solution. The mixture was shaken vigorously on an IKA VORTEX2 vortex agitator (Janke & Kunkel, Staufen, Germany) at 2800 rpm for 60 s. The emulsion was then divided into two phases by centrifugation at 8000 rpm for 5 min at room temperature, and the sample to be analyzed was that below the CH_2Cl_2 phase, which was extracted for GC-MS and GC-O-MS analysis.

2.4. Identification by GC-MS and GC-O-MS

GC-MS analysis was performed on an Agilent 7890B gas chromatograph equipped with an Agilent 5977A mass selective detector (MSD) (Agilent Technologies, USA). All samples were analyzed on a DB-WAX column (60 m × 0.25 mm i.d., 0.25 μm film thick; J&W Scientific, USA). The flow of carrier gas (helium) was set to 1 mL/min, and the injection (1 μL) was performed in splitless mode at 250 °C. The oven temperature increased from 40° to 50°C (held for 5 min) at a rate of 10 °C/min, then increased from 50° to 80°C (held at 5 min) at a rate of 3 °C/min, and finally increased at a rate of 5 °C/min until it reached 230 °C (10 min).

GC-O-MS analysis was performed on an Agilent 7890B gas chromatograph equipped with an Agilent 5977A MSD and a sniff port (ODP3, Gerstel, Germany). All samples were analyzed with a DB-FFAP column (60 m × 0.25 mm i.d., 0.25 μm film thick; J&W Scientific, USA) and an HP-5MS column (60 m × 0.25 mm i.d., 0.25 μm film thick; J&W Scientific, USA). The temperature of the olfactory port was 250 °C, the full-scan range was *m/z* 35–400, the ion source temperature was 230 °C and the other GC conditions were the same as those for GC-MS. Three trained panelists (3 females, aged 22–24) from the Laboratory of Food, College of Applied Chemistry at Beijing Technology and Business University were chosen to perform the GC-O-MS analysis. During the GC

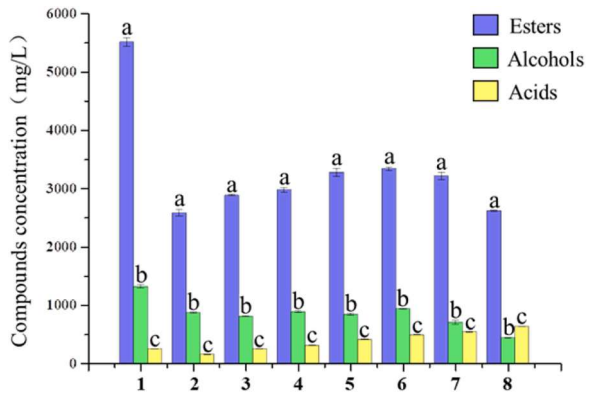


Fig. 1. Variation in total ester, alcohol and acid concentrations of raw Baijiu from distillation processes. The concentrations of volatile compounds were represented as the mean and standard deviation (mean ± SD, n = 3). Letters a, b and c represents significant variation among total ester, alcohol and acid from raw Baijiu (one-way ANOVA with Bonferroni test, $P < 0.05$).

operation, an evaluator put her nose close to the sniffing port and recorded the odor, intensity and retention time of the stimulus. The aroma intensities of the volatile compounds were scored from 0 to 5, "0" was none, "3" was moderate, and "5" was extreme. All analyses were repeated in triplicate by each panelist. Before the GC-O analysis, panelists were trained by sniffing 50 reference compounds in their concentrations 5 times above their odor thresholds in water or air (Jia et al., 2020; Xu et al., 2022; Li et al., 2019; Wu et al., 2021).

The aroma compounds were identified by comparing the mass spectrum, aroma and linear retention index (LRI) with those of the pure standards. The LRI was calculated by injecting the C_7–C_{40} *n*-alkane mixture under the same GC-MS conditions as those for Baijiu samples.

1-Hexanol, ethyl acetate, ethyl hexanoate, and ethyl lactate were quantified by gas chromatography with a flame ionization detector (GC-FID) because these compounds were present in high concentrations. Their separations were performed on a DB-FFAP column and the other GC conditions were the same as those for GC-MS: one microliter of Baijiu sample with 40 mg/L spiked pentyl acetate and 4-octanol was injected into the GC in splitless mode. Stock solutions were prepared by mixing the standard and diluting it with absolute ethanol, followed by 2-fold serial dilutions with 52% ethanol. Then, IS1 and IS2 were added to the standard solution, which was analyzed by GC-FID in the same way as the Baijiu sample. The standard curve was constructed by taking the ratio of the concentrations of the target compound to the internal standards as the abscissa and the ratio of the responses of the target compound to the internal standards as the ordinate. The concentrations of the target compounds in the Baijiu sample were calculated according to the calibration curves. A correction factor was estimated from the ratio of the response of the target compound to the internal standard compound at the same concentration. Other compounds were quantified by GC-MS using the internal standard method, and the analysis conditions were the same as those for the GC-MS analysis described above.

2.5. HS-GC-IMS analysis

HS-GC-IMS analysis was performed on an advanced IMS system manufactured by FlavourSpec® (G.A.S., Dortmund, Germany) coupled with an Agilent 7890B gas chromatograph. The system was equipped with an Acrichi Automatic Headspace Sampler (Beijing Juxinzhuifeng Technology Co., Ltd., China) as the headspace sampling unit.

The sample preparation and instrumental analysis were performed according to the methods reported by He et al. (He et al., 2021). An appropriate amount of Baijiu sample was diluted to below 30% by the same multiple with ultrapure water. For volatile organic compound (VOC) analysis, 1 mL of diluted Baijiu sample was put into a 20 mL headspace vial and subsequently incubated at 60 °C for 10 min. The relevant parameter settings of the instrument were consistent with the above method.

All analyses were performed in triplicate. Volatile compounds were identified by comparing the LRI and drift time to standards in the GC-IMS library.

2.6. Calculation of the odor activity values (OAVs)

The OAVs of the aroma compounds were calculated from the ratio of the concentrations detected in the Baijiu samples compared to their threshold values (Grosch, 1994). The threshold values were obtained by searching relevant literature (Fan et al., 2015; Gemert, 2003; Gao et al., 2014; Zhao et al., 2018; Zheng et al., 2016; Sha, 2017; Wang et al., 2014). Odor thresholds of most aromas were determined in 46% ethanol/water solution and aromas determined in other media had been noted below in Table 5.

2.7. Statistical analysis

Statistical analysis was performed through one-way analysis of variance (ANOVA) to identify significant differences among raw Baijiu samples from different distillation stages using Origin version 9.0 software (OriginLab Co., Massachusetts, USA). The HS-GC-IMS data were viewed and processed by LAV software (G.A.S., Dortmund, Germany). Heatmap analysis and hierarchical cluster analysis (HCA) were conducted by MetaboAnalyst5.0. PCA and PLS-DA were carried out with SIMCA version 14.0 (Umetrics, Umea, Sweden). ROC was carried out with SPSS version 16.0 (SPSS Inc., USA).

3. Results and discussion

3.1. Volatile compounds of raw Baijiu at different distillation stages

During Baijiu distillation, its temperature will continue to increase under the continuous heating of steam. Many trace aroma compounds in the fermented grains will also enter the raw Baijiu body with the steam during the distillation. Due to the differences in the numbers and contents of the aroma compounds, raw Baijiu had different aroma characteristics in different distilling stages.

First, the variation of the ethanol content was analyzed and the results were shown in Table 1. The ethanol content decreased slowly in the first stage, and then decreased rapidly in the later stage of the distillation in the process of raw Baijiu distillation, and the overall trend was gradually decreasing. The highest ethanol content was 80%, and the lowest ethanol content was 31%.

Then, a total of 140 volatile compounds were detected in 8 raw Baijiu samples during distillation by GC-MS, including 74 esters, 19 alcohols, 15 acids, 3 aldehydes, 5 ketones, 4 acetal compounds, 6 nitrogenous compounds, 2 heterocyclic compounds, 7 phenols and 5 other compounds. Among them, the esters, alcohols, and acids were the most diverse and highest-content volatile aroma substances detected. Therefore, this experiment mainly studied the changes in esters, alcohols (except ethanol), and acids that greatly contribute to the flavor of Baijiu.

The total concentrations of esters, alcohols and acids from the distillation process were quantified and the results are shown in Fig. 1. From Fig. 1, we can see that the content of esters in each sample were significantly higher than those of the alcohols and acids ($p < 0.05$). During the whole distillation process, the content of total esters decreased first, then slowly increased, and finally decreased again. The content of total esters of the first sample at the beginning of the distillation was significantly higher than the other samples, and the last sample was significantly lower than the other samples. Therefore, the content of total esters decreased during the distillation process. The reason for this trend was that a large amount of ethanol in the fermented grains was steamed out at the beginning of the distillation, and the

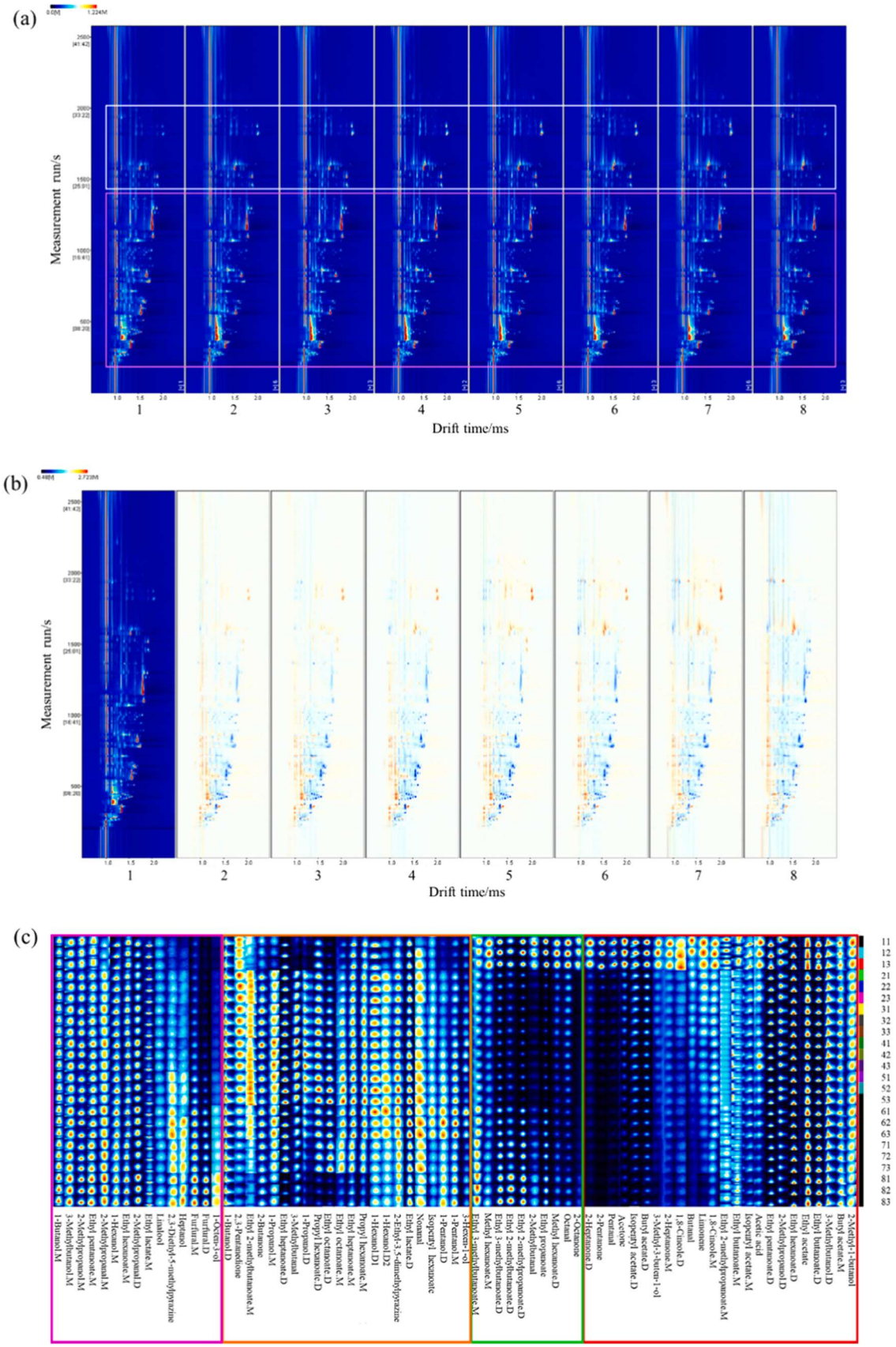


Fig. 2. HS-GC-IMS analysis based on raw Baijiu: (a) 2D topographic images of raw Baijiu samples; (b) 2D topographic images with differential comparison mode of raw Baijiu samples; and (c) fingerprints of all samples generated using the Gallery Plot.

Table 2
Information on the Qualitative Substances (71 peaks for 47 compounds).

No.	Compound	LRI[b]	Rt[c] (×10³/s)	Dt[d] (RIPrel)	Comment
1	Linalool	1520	2.22	1.25	
2	1-Octen-3-ol	1472	2.03	1.15	
3	2,3-Diethyl-5-methylpyrazine	1472	2.03	1.27	
4	Furfural	1451	1.95	1.08	Monomer
5	Furfural	1450	1.95	1.34	Dimer
6	Acetic acid	1448	1.94	1.14	
7	2-Ethyl-3,5-dimethylpyrazine	1451	1.95	1.21	
8	1-Heptanol	1445	1.92	1.39	
9	Isopentyl hexanoate	1447	1.93	1.54	
10	Ethyl octanoate	1433	1.88	1.47	Monomer
11	Ethyl octanoate	1432	1.88	2.03	Dimer
12	1-Hexanol	1367	1.62	1.32	Monomer
13	1-Hexanol	1367	1.62	1.64	Dimer
14	1-Hexanol	1365	1.61	2.00	Dimer
15	Ethyl lactate	1357	1.58	1.14	Monomer
16	Ethyl lactate	1356	1.57	1.54	Dimer
17	1-Nonanal	1357	1.58	1.48	
18	(Z)− 3-Hexen-1-ol	1344	1.53	1.25	
19	Propyl hexanoate	1328	1.46	1.39	Monomer
20	Propyl hexanoate	1327	1.46	1.92	Dimer
21	Ethyl heptanoate	1343	1.53	1.40	Monomer
22	Ethyl heptanoate	1341	1.52	1.92	Dimer
23	Octanal	1302	1.36	1.41	
24	Ethyl hexanoate	1246	1.16	1.33	Monomer
25	Ethyl hexanoate	1248	1.17	1.80	Dimer
26	1-Pentanol	1263	1.22	1.25	Monomer
27	1-Pentanol	1263	1.22	1.51	Dimer
28	3-Methylbutanol	1221	1.07	1.24	Monomer
29	3-Methylbutanol	1220	1.07	1.49	Dimer
30	2-Methyl-1-butanol	1218	1.06	1.23	
31	3-Methyl-3-buten-1-ol	1205	1.02	1.17	
32	DL-Limonene	1205	1.02	1.30	
33	Methyl hexanoate	1197	0.991	1.29	Monomer
34	Methyl hexanoate	1196	0.989	1.68	Dimer
35	2-Heptanone	1192	0.974	1.26	Monomer
36	2-Heptanone	1192	0.973	1.64	Dimer
37	1,8-Cineole	1201	1.00	1.33	Monomer
38	1,8-Cineole	1200	1.00	1.74	Dimer
39	Butanol	1158	0.863	1.18	Monomer
40	Butanol	1159	0.866	1.38	Dimer
41	Ethyl pentanoate	1146	0.823	1.27	Monomer
42	Ethyl pentanoate	1145	0.821	1.68	Dimer
43	Isopentyl acetate	1133	0.782	1.31	Monomer
44	Isopentyl acetate	1134	0.783	1.75	Dimer
45	2-Methylpropanol	1112	0.713	1.17	Monomer
46	2-Methylpropanol	1113	0.715	1.36	Dimer
47	Butyl acetate	1088	0.645	1.24	Monomer
48	Butyl acetate	1088	0.644	1.63	Dimer
49	Ethyl 3-methylbutanoate	1082	0.632	1.25	Monomer
50	Ethyl 3-methylbutanoate	1083	0.633	1.66	Dimer
51	Ethyl butanoate	1049	0.559	1.21	Monomer
52	Ethyl butanoate	1048	0.558	1.56	Dimer
53	Ethyl 2-methylbutanoate	1064	0.592	1.24	Monomer
54	Ethyl 2-methylbutanoate	1064	0.593	1.66	Dimer
55	1-Propanol	1055	0.573	1.11	Monomer
56	1-Propanol	1056	0.574	1.25	Dimer
57	2,3-Pentanedione	1039	0.538	1.22	
58	2-Pentanone	997	0.447	1.38	
59	Pentanal	996	0.444	1.43	
60	Ethyl propanoate	974	0.417	1.46	
61	Ethyl 2-methylproanoate	981	0.425	1.21	Monomer
62	Ethyl 2-methylproanoate	981	0.425	1.57	Dimer
63	2-Octanone	967	0.408	1.35	
64	Ethyl acetate	910	0.339	1.34	
65	Butanal	915	0.344	1.29	
66	3-Methylbutanal	928	0.361	1.41	
67	2-Butanone	867	0.286	1.06	
68	Acetone	860	0.277	1.12	
69	2-Methylpropanal	856	0.273	1.10	Monomer
70	2-Methylpropanal	856	0.272	1.29	Dimer
71	2-Methylbutanal	973	0.416	1.40	

[a]MW: molecular weight
[b]LRI: linear retention index
[c]Rt: retention time
[d]Dt: drift time

alcohol-soluble esters were steamed out together. Through research, it was also confirmed that there were a lot of alcohol-soluble and small molecular weight esters in the raw Baijiu at the beginning of the distillation.

The changes in the content of alcohols were similar to those of the esters. The content of total alcohols gradually decreased with the extension of distillation time. The content of total alcohols of the first sample at the beginning of the distillation was significantly higher than the other samples, and the last sample was significantly lower than the other samples. This result echoed the result of the variation in the alcohol content. Most alcohols were small molecular weights and low boiling points and were largely distilled out with ethanol in the initial stage of distillation. As the alcohols in the fermented grains were gradually distilled out, the alcohols content in the Baijiu samples gradually decreased in the later distillation process.

Total acids showed an upward trend overall with the extension of the distillation. The acid content was low in the early stage of distillation and reached their maximum in the last Baijiu sample. Acid compounds were mostly concentrated in the late distillations because most organic acids were easily soluble in water but hardly soluble in ethanol, and their boiling points were high.

These results were consistent with the variations of volatile aroma substances during the distillation process of strong flavor Baijiu reported by predecessors (Li et al., 2012; Zhang et al., 2018; Xu et al., 2019; He et al., 2021).

3.2. *The analysis of the differences of raw Baijiu at different distillation stages by HS-GC-IMS*

The volatile components of different distilling stages with different layers of fermented grains were detected via HS-GC-IMS analysis.

A FlavourSpec® instrument was used to generate data, as shown in Fig. 2(a), (b) and (c). Fig. 2(a) showed 2D topographic plots, which compared the differences in the content and kinds of compounds between the raw Baijiu samples during the entire distillation process. The background of all samples was blue, and the red vertical line at the horizontal coordinate of 1.0 was the reactive ion peak (RIP), which was normalized. Each point on the right side of the RIP symbolized a VOC. Then, to compare the differences among samples from different samples, a differential comparisons mode could be adopted, as seen in Fig. 2(b). That was, the spectral diagram of one sample (the first Baijiu sample) was selected as the reference, while the spectral diagram of other samples was deducted from the reference. If two VOCs were identical, then the background after deduction was white. Red spots indicate that the concentration of the substance was higher than that of the reference, whereas blue spots indicate that the concentration of the substance was lower than that of the reference. In addition, a difference comparison of fingerprint maps that involved all plots and areas of the compounds at different samples was shown in Fig. 2(c), where each row represents a sample and each column represents a compound; each sample was analyzed twice in parallel. By drawing fingerprints, we could compare the content of each compound in each sample and its changing trend throughout the distillation process. A total of 71 signal peaks were identified and marked with an existing name, as presented in Table 2.

As shown in Fig. 2(a), the compound content in the white box gradually increased with the extension of the distillation time, while the trend of the compounds in the pink box was the opposite, that was, they showed a gradually decreasing trend with the extension of the distillation time.

Additionally, as shown in Fig. 2(b), with the extension of the distillation time, the peaks of most compounds gradually deepened in blue,

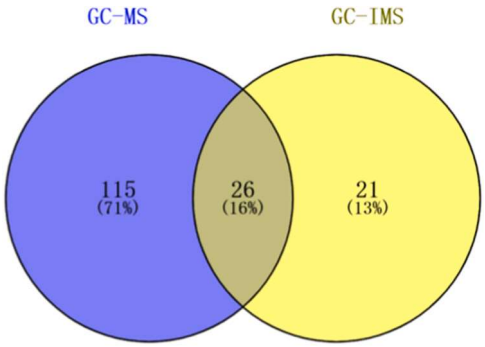


Fig. 3. Venn diagram of volatile compounds via HS-GC-IMS and GC-MS.

that was, these compounds' content gradually decreased. Through research, it was found that these compounds were mostly low molecular weight and alcohol-soluble compounds, which appeared first in the two-dimensional spectrum; some peaks of compounds gradually deepened in red, that was, these compounds' content gradually increased, and a few material peaks were gradually deepened in red and then disappeared, that was, these compounds' content first increased and then decreased. Most of these compounds had a large molecular weight and they were not easily soluble in alcohol, so they appeared in the later part of the two-dimensional spectrum.

Combined with the fingerprint spectrum in Fig. 2(c), we can conclude that: (1) the concentrations of 2-methyl-1-butanol, 3-methyl-butanol, ethyl butanoate, ethyl acetate, ethyl hexanoate, 2-methylpropanol, ethyl pentanoate, acetic acid, limonene, butanal, 1,8-cineole, 3-methyl-3-buten-1-ol, butyl acetate, isopentyl acetate, acetone, pentanal, 2-pentanone, and 2-heptanone in the red frame gradually decreased with increasing distillation time. These compounds had small molecular weight and were volatile or easily soluble in ethanol. They were usually concentrated in the early stages of distillation; (2) the concentrations of 2-octanone, octanal, methyl hexanoate, ethyl propanoate, 2-methylbutanal, ethyl 2-methylpropanoate, ethyl 2-methylbutanoate, and ethyl 3-methylbutanoate in the green frames showed a tendency to decrease and then increase with the extension of distillation time. These compounds were all soluble in ethanol, slightly soluble or insoluble in water, and entered the Baijiu body through ethanol extraction in the early stage of distillation. The ethanol concentration was the highest at the beginning of distillation, so these compounds entered the Baijiu body with ethanol in large quantities. As the distillation progresses, the ethanol concentration decreased, and the content of these compounds also decreased; as the distillation continued, the temperature continued to rise and gradually reached the boiling point of these compounds, hence, these compounds continued to be steamed out and their content in the Baijiu body also gradually increased. Therefore, these compounds showed a tendency to decrease and then increase in the whole distillation process; (3) the concentrations of 3-hexen-1-ol, 1-pentanol, isoamyl hexanoate, nonanal, ethyl lactate, 2-ethyl-3,5-dimethylpyrazine, 1-hexanol, ethyl octanoate, propyl hexanoate, 1-propanol, 3-methyl-butanal, ethyl heptanoate, 2-butanone, 2,3-pentanedione, and 1-butanol in the orange frames showed a tendency to increase and then decrease with the extension of distillation time. The boiling point of these compounds was slightly higher or the volatility was slightly lower. The temperature was low in the early stage and they were not steamed out. In the later stage, they were slowly steamed out as the temperature increased until they were completely steamed out at the end of the distillation; (4) the concentrations of 1-octen-3-ol, furfural, heptanol, 2,3-diethyl-5-methylpyrazine, linalool, and 2-methylpropanal in the pink frames gradually increased with increasing distillation time. These compounds had high boiling points and were easily soluble in water. They accumulated in the post-distillation stage.

Table 3
Aroma compounds identified by gas chromatography−olfactometry-mass spectrometry in Baijiu samples.

No.	Aroma compounds[a]	Descriptor[b]	LRI (DB-WAX[c])	Basis of ID[d]
1	Ethyl butanoate	pineapple	1041	MS, aroma, RI, S
2	Butyl acetate	pear	1080	MS, aroma, RI, S
3	2-Methylpropanol	ethereal winey cortex	1094	MS, aroma, RI, S
4	Ethyl pentanoate	apple	1143	MS, aroma, RI, S
5	1-Butanol	alcoholic	1150	MS, aroma, RI, S
6	Ethyl 4-methylpentanoate	fruity	1243	MS, aroma, RI, S
7	3-Methyl-1-butanol	fusel oil	1249	MS, aroma, RI, S
8	Ethyl hexanoate	fruity	1267	MS, aroma, RI, S
9	Hexyl acetate	fruity	1289	MS, aroma, RI, S
10	Furfuryl ethyl ether	sweet spicy	1299	MS, aroma, RI, S
11	Cyclopentanol	fusel oil	1312	MS, aroma, RI, S
12	Propyl hexanoate	sweet fruity	1344	MS, aroma, RI, S
13	Ethyl lactate	fruity	1362	MS, aroma, RI, S
14	1-Hexanol	green	1367	MS, aroma, RI, S
15	Ethyl 2-hydroxybutanoate	rotten	1416	MS, aroma, RI, S
16	Hexyl butanoate	green waxy soapy	1423	MS, aroma, RI, S
17	Propyl lactate	fruity	1433	MS, aroma, RI, S
18	Ethyl octanoate	fruity	1439	MS, aroma, RI, S
19	Acetic acid	vinegar	1453	MS, aroma, RI, S
20	Isopentyl hexanoate	fruity	1459	MS, aroma, RI, S
21	Furfural	almond woody	1470	MS, aroma, RI, S
22	2-Acetylfuran	cocoa bitter	1501	MS, aroma, RI, S
23	Furfuryl acetate	fruity bitter	1529	MS, aroma, RI, S
24	Ethyl nonanoate	sweet fruity	1523	MS, aroma, RI, S

(*continued on next page*)

Table 3 (*continued*)

No.	Aroma compounds[a]	Descriptor[b]	LRI (DB-WAX[c])	Basis of ID[d]
25	Ethyl DL-2-hydroxy-4-methylpentanoate	sweet fruity	1534	MS, aroma, RI, S
26	Hexyl hexanoate	sweet fruity	1596	MS, aroma, RI, S
27	Butanoic acid	sweaty rancid	1611	MS, aroma, RI, S
28	Ethyl decanoate	fruity waxy	1624	MS, aroma, RI, S
29	Phenylacetaldehyde	floral	1645	MS, aroma, RI, S
30	2-Furanmethanol	musty	1653	MS, aroma, RI, S
31	3-Methylbutanoic acid	sweaty	1655	MS, aroma, RI, S
32	Diethyl butanedioate	fruity	1668	MS, aroma, RI, S
33	2-Methoxyhexyl acetate	sweet mushroom	1712	MS, aroma, RI, S
34	3-Methylthiopropanol	onion mushroom	1721	MS, aroma, RI, S
35	Pentanoic acid	sweaty	1726	MS, aroma, RI, S
36	Hexanoic anhydride	rotten	1759	MS, aroma, RI, S
37	Diethyl glutarate	Woody	1775	MS, aroma, RI, S
38	Ethyl phenylacetate	floral sweet	1788	MS, aroma, RI, S
39	4-Methylpentanoic acid	rancid	1794	MS, aroma, RI, S
40	Phenethyl acetate	floral rose	1821	MS, aroma, RI, S
41	Hexanoic acid	sweaty rancid	1839	MS, aroma, RI, S
42	Ethyl 3-phenylpropanoate	floral	1898	MS, aroma, RI, S
43	2-Phenylethanol	rosy honey	1924	MS, aroma, RI, S
44	Heptanoic acid	rancid	1956	MS, aroma, RI, S
45	2-Methoxy-*p*-cresol	spice phenolic	1976	MS, aroma, RI, S
46	Phenol	phenolic plastic rubber	2021	MS, aroma, RI, S
47	4-Ethyl-2-methoxyphenol	phenolic clove	2052	MS, aroma, RI, S
48	*p*-Cresol	phenolic animal	2101	MS, aroma, RI, S
49	Methyl palmitate	waxy bitter	2227	MS, aroma, RI, S
50	Ethyl linolenate	milky sweet roasty	2613	MS, aroma, RI, S

[a] Aroma active compounds;
[b] Odor quality perceived at the sniffing port;
[c] DB-WAX, linear retention index on DB-WAX;
[d] Identification based on LRI (linear retention index), MS (mass spectrometry), aroma (odor description by comparison to the pure standard) or S (standard).

3.3. Comparison of the results via HS-GC-IMS and GC-MS

Table 2 indicates that in 8 raw Baijiu samples, 47 compounds were identified via HS-GC-IMS, and 140 ones were identified via GC-MS, respectively. Also, 26 mutual compounds had been identified via these two types of analysis methods as shown in Fig. 3.

Some compounds, including 2-methyl-1-butanol, ethyl acetate, limonene, butanal, 1,8-cineole, 3-methyl-3-buten-1-ol, 2-propanone, pentanal, 2-octanone, octanal, 2-methylbutanal, 3-hexen-1-ol, nonanal, 2-ethyl-3,5-dimethylpyrazine, 3-methylbutanal, 2-butanone, 2,3-pentanedione, 1-octen-3-ol, 2,3-diethyl-5-methylpyrazine, linalool, and 2-methylpropanal could not be detected by GC-MS, but they were detected by HS-GC-IMS. These compounds were mostly aldehydes and ketones, and GC-IMS was more sensitive to the response of these compounds, so it was more suitable for the detection of aldehydes and ketones. Due to the high sensitivity of GC-IMS, the compounds could be separated by GC column and chemical ions characterized substances based on their differences in the rate of migration of the gas phase ions in an electric field. Although Fig. 2 showed the changing trends of compounds in different samples, some volatile compounds could not be confirmed due to the incomplete database. GC-MS identified more compounds than HS-GC-IMS, but the sample must be pretreated and extracted via LLME for GC-MS analysis. The detection efficiency of GC-MS would decrease without pretreatment. Therefore, the combination of GC-IMS and GC-MS allows for more comprehensive identification and quantification of volatile compounds.

3.4. Qualitative analysis of aroma compounds via GC-O-MS

To identify compounds that contribute to the overall aroma profile, GC-O-MS was used to analyze the raw Baijiu samples. A total of 50 odorants (Table 3) were identified by mass spectrometry, aroma and LRIs, including 26 esters, 8 alcohols, 8 acids, 2 aldehydes, 4 aromatic compounds, 1 heterocyclic compound, and 1 other compound. Table 4 shows the concentrations of the compounds in each sample. As shown in Table 5, the OAVs of these 50 aroma compounds ranged from 0 to 46521 according to GC-O-MS analysis. Based on Table 5, ethyl hexanoate contributed a strong fruity flavor to the aroma of Baijiu and had the highest OAV (7653–46521) among the aroma active compounds and reached its maximum in the first Baijiu sample. In addition, ethyl butanoate (233–11465; pineapple), ethyl pentanoate (288–7145; apple), and ethyl octanoate (298–3934; fruity) had high OAV values. These compounds basically showed a gradually decreasing trend with the extension of the distillation time, followed by ethyl 4-methylpentanoate (fruity), butyl acetate (pear), 1-butanol (alcoholic), and these compounds also gradually decreased with the extension of the distillation time. 4-Ethyl-2-methoxyphenol (phenolic clove), butanoic acid (sweaty, rancid), hexanoic acid (sweaty, rancid), 1-hexanol (green), ethyl 3-phenylpropanoate (floral), pentanoic acid (sweaty), and *p*-cresol (phenolic animal) had a contrary change trend to the above substances, and it gradually increased with the extension of the distillation time.

Table 4
Concentrations of 50 aroma compounds in 8 raw Baijiu samples.

No.	Aroma compounds	Concentration (mg/L)							
		1	2	3	4	5	6	7	8
1	Ethyl butanoate	940 ± 14.6	206 ± 3.2	138 ± 0.727	113 ± 2.27	73.9 ± 1.28	57 ± 0.543	20.3 ± 0.31	19.1 ± 0.17
2	Butyl acetate	47.1 ± 0.253	6.42 ± 0.164	4.17 ± 0.017	2.92 ± 0.009	2.07 ± 0.02	1.39 ± 0.012	0.625 ± 0.012	0.431 ± 0.008
3	2-Methylpropanol	136 ± 2.98	42.6 ± 0.37	29.8 ± 0.398	25.2 ± 0.526	20.8 ± 0.281	17.3 ± 0.177	10.4 ± 0.061	7.14 ± 0.018
4	Ethyl pentanoate	193 ± 0.095	46.4 ± 1.05	34.1 ± 0.444	25.1 ± 0.096	21 ± 0.379	15.3 ± 0.265	11.9 ± 0.144	7.77 ± 0.15
5	1-Butanol	511 ± 11	335 ± 4.22	299 ± 4.4	309 ± 7.15	270 ± 6.35	267 ± 3.77	175 ± 0.6	124 ± 0.564
6	Ethyl 4-methylpentanoate	4.36 ± 0.068	1.57 ± 0.014	1.52 ± 0.013	1.2 ± 0.016	0.57 ± 0.165	0.643 ± 0.053	0.55 ± 0.296	——
7	3-Methyl-1-butanol	372 ± 6.31	226 ± 1.79	196 ± 2.28	202 ± 3.49	188 ± 3.37	170 ± 0.723	118 ± 0.112	79.1 ± 0.358
8	Ethyl hexanoate[a]	2559 ± 32.9	1157 ± 20	1092 ± 2.99	879 ± 6.12	861 ± 13.5	679 ± 2.38	714 ± 3.38	421 ± 1.36
9	Hexyl acetate	21.2 ± 0.02	9.13 ± 0.049	9.2 ± 0.057	6.94 ± 0.105	10.5 ± 0.136	1.39 ± 0.205	0.779 ± 0.126	0.297 ± 0.027
10	Furfuryl ethyl ether	42.9 ± 0.029	13.9 ± 0.182	9.26 ± 0.054	7.72 ± 0.052	6 ± 0.021	4 ± 0.005	4.04 ± 0.055	2.5 ± 0.01
11	Cyclopentanol	——	——	0.349 ± 0.014	0.431 ± 0.026	——	0.67 ± 0.038	0.987 ± 0.027	1.2 ± 0.021
12	Propyl hexanoate	20.7 ± 0.112	11.6 ± 0.217	13.2 ± 0.084	10.9 ± 0.105	14.1 ± 0.162	4.87 ± 0.428	6.01 ± 0.12	——
13	Ethyl lactate[a]	1159 ± 19.5	869 ± 2.76	1151 ± 3.68	1519 ± 20.4	1543 ± 33	1869 ± 15.2	1655 ± 46.7	1585 ± 2.82
14	1-Hexanol[a]	164 ± 2.34	206 ± 1.55	230 ± 1.61	288 ± 3.08	286 ± 6.56	373 ± 1.57	274 ± 12.7	124 ± 2.34
15	Ethyl 2-hydroxybutanoate	0.318 ± 0.006	0.502 ± 0.013	0.855 ± 0.01	1.24 ± 0.005	1.53 ± 0.033	2.67 ± 0.01	3.33 ± 0.008	3.34 ± 0.024
16	Hexyl butanoate	17.2 ± 0.24	7.91 ± 0.137	11.2 ± 0.122	5.67 ± 0.021	12.2 ± 0.09	5.55 ± 0.053	10.9 ± 0.056	1.03 ± 0.015
17	Propyl lactate	2.48 ± 0.002	3.32 ± 0.056	5.58 ± 0.069	8.56 ± 0.078	11.6 ± 0.054	19.8 ± 0.076	16.3 ± 0.111	22.2 ± 0.03
18	Ethyl octanoate	51.1 ± 0.181	22.8 ± 0.45	34.7 ± 0.263	19.6 ± 0.135	41.4 ± 0.293	20.5 ± 0.134	38.4 ± 0.376	3.88 ± 0.013
19	Acetic acid	51.5 ± 0.458	44.9 ± 0.597	60.6 ± 0.575	74.9 ± 0.933	97.1 ± 0.515	62.1 ± 0.897	124 ± 0.433	148 ± 0.288
20	Isopentyl hexanoate	5.73 ± 0.15	1.94 ± 0.168	4.26 ± 0.071	2.57 ± 0.044	7.01 ± 0.147	2.39 ± 0.097	5.51 ± 0.171	——
21	Furfural	16.8 ± 0.066	11.6 ± 0.16	24.4 ± 0.018	42.7 ± 0.241	60.4 ± 0.337	110 ± 0.908	123 ± 0.569	129 ± 0.792
22	2-Acetylfuran	——	——	——	——	——	0.651 ± 0.035	0.516 ± 0.107	0.956 ± 0.077
23	Furfuryl acetate	0.022 ± 0.004	0.72 ± 0.069	1.2 ± 0.06	1.4 ± 0.194	0.629 ± 0.13	0.852 ± 0.083	1.39 ± 0.291	1.54 ± 0.058
24	Ethyl nonanoate	1.07 ± 0.094	——	0.548 ± 0.097	——	——	——	——	——
25	Ethyl DL-2-hydroxy-4-methylpentanoate	21.2 ± 0.159	61.6 ± 3.13	99.4 ± 2.16	144 ± 4.63	186 ± 6.12	279 ± 4.51	276 ± 3.4	225 ± 0.989
26	Hexyl hexanoate	26.5 ± 0.33	12.1 ± 0.246	26.3 ± 0.281	13.6 ± 0.12	44 ± 0.216	21 ± 0.221	33.8 ± 0.33	0.688 ± 0.005
27	Butanoic acid	65.3 ± 0.731	40.4 ± 0.349	64.9 ± 0.428	78.9 ± 0.554	103 ± 0.991	153 ± 0.446	159 ± 0.52	188 ± 1.21
28	Ethyl decanoate	4 ± 0.085	1.1 ± 0.064	3.42 ± 0.06	2.42 ± 0.033	8.14 ± 0.339	1.08 ± 0.097	8.66 ± 0.08	——
29	Phenylacetaldehyde	2.01 ± 0.042	1.9 ± 0.019	3.38 ± 0.154	5.52 ± 0.107	6.73 ± 0.472	13 ± 0.343	16.5 ± 0.106	18.8 ± 0.112
30	2-Furanmethanol	6.74 ± 0.035	4.23 ± 0.056	5.51 ± 0.1	7.66 ± 0.054	9.89 ± 0.16	17.1 ± 0.081	16.7 ± 0.247	21.4 ± 0.091
31	3-Methylbutanoic acid	4.93 ± 0.02	2.5 ± 0.009	4.71 ± 0.041	6.34 ± 0.025	12.3 ± 0.531	18.2 ± 0.141	18.8 ± 0.061	25.5 ± 0.083
32	Diethyl butanedioate	20.3 ± 0.084	11.4 ± 0.101	22.4 ± 0.2	36.4 ± 0.276	50.4 ± 0.184	82.5 ± 0.959	94.4 ± 0.703	94.6 ± 0.441
33	2-Methoxyhexyl acetate	2.38 ± 0.162	2.06 ± 0.049	3.87 ± 0.049	5.88 ± 0.101	9.74 ± 0.131	16.3 ± 0.139	22.4 ± 0.189	21.5 ± 0.074
34	3-Methylthiopropanol	0.206 ± 0.009	0.148 ± 0.025	0.142 ± 0.031	0.211 ± 0.002	0.277 ± 0.026	0.261 ± 0.021	0.293 ± 0.02	0.352 ± 0.003
35	Pentanoic acid	6.49 ± 0.063	2.91 ± 0.021	5.46 ± 0.003	7.43 ± 0.031	11.2 ± 0.115	18 ± 0.173	21.6 ± 0.459	29.5 ± 0.226
36	Hexanoic anhydride	——	0.502 ± 0.031	0.327 ± 0.035	1.13 ± 0.012	2.46 ± 0.031	3.04 ± 0.05	6 ± 0.017	5.57 ± 0.051
37	Diethyl glutarate	——	——	——	0.143 ± 0.015	——	1.14 ± 0.094	0.514 ± 0.028	0.6 ± 0.005
38	Ethyl phenylacetate	2.05 ± 0.022	4.73 ± 0.069	7.54 ± 0.086	11.3 ± 0.101	21 ± 0.118	29.1 ± 0.226	33.5 ± 0.23	29.6 ± 0.107
39	4-Methylpentanoic acid	——	——	——	0.412 ± 0.026	0.393 ± 0.025	0.525 ± 0.002	0.675 ± 0.015	0.843 ± 0.031

(*continued on next page*)

Table 4 (*continued*)

No.	Aroma compounds	Concentration (mg/L)							
		1	2	3	4	5	6	7	8
40	Phenethyl acetate	0.306 ± 0.02	——	1.02 ± 0.017	1.56 ± 0.024	2.01 ± 0.067	3.4 ± 0.695	4.97 ± 0.069	5.04 ± 0.099
41	Hexanoic acid	113 ± 1.25	57.1 ± 0.362	103 ± 0.507	130 ± 1.32	175 ± 3.9	221 ± 2.54	187 ± 0.701	200 ± 1.41
42	Ethyl 3-phenylpropanoate	2.72 ± 0.139	2.63 ± 0.158	4.13 ± 0.137	5.11 ± 0.121	12 ± 0.272	13.9 ± 0.06	20.4 ± 0.155	17.2 ± 0.062
43	2-Phenylethanol	4.84 ± 0.017	2.15 ± 0.071	3.22 ± 0.018	5.13 ± 0.022	7.73 ± 0.148	13.5 ± 0.12	13.1 ± 0.127	17.6 ± 0.03
44	Heptanoic acid	0.593 ± 0.022	0.246 ± 0.072	0.767 ± 0.038	1.15 ± 0.01	1.6 ± 0.046	3.16 ± 0.085	4.59 ± 0.04	6.84 ± 0.031
45	2-Methoxy-*p*-cresol	——	——	0.125 ± 0.005	0.2 ± 0.02	0.215 ± 0.015	0.279 ± 0.052	——	——
46	Phenol	0.879 ± 0.012	0.442 ± 0.043	0.329 ± 0.021	0.447 ± 0.014	——	0.733 ± 0	0.719 ± 0.004	1.07 ± 0.058
47	4-Ethyl-2-methoxyphenol	0.791 ± 0.072	0.448 ± 0.02	0.633 ± 0.014	1.17 ± 0.012	1.79 ± 0.104	2.99 ± 0.078	3.51 ± 0.086	4.14 ± 0.02
48	*p*-Cresol	0.474 ± 0.052	0.391 ± 0.024	0.446 ± 0.003	0.645 ± 0.023	10.9 ± 0.589	1.52 ± 0.104	1.75 ± 0.052	1.98 ± 0.009
49	Methyl palmitate	——	——	——	——	3.18 ± 0.055	3.18 ± 0.131	2.15 ± 0.039	1.67 ± 0.039
50	Ethyl linolenate	1.75 ± 0.013	0.597 ± 0.005	1.01 ± 0.038	0.588 ± 0.041	——	——	——	——

[a] Aroma compounds quantified by GC–FID

Table 5
Odor activity values of 40 aroma compounds in 8 raw Baijiu samples.

No.	Aroma compounds	Odor threshold (μg/L)	OAV							
			1	2	3	4	5	6	7	8
1	Ethyl butanoate	82[a]	11,465	2515	1677	1373	902	695	247	233
2	Butyl acetate	58[b*]	811	111	72	50	36	24	11	7
3	2-Methylpropanol	28,300[c]	5	2	1	< 1	< 1	< 1	< 1	< 1
4	Ethyl pentanoate	27[a]	7145	1720	1262	929	777	567	439	288
5	1-Butanol	2730[d]	187	123	109	113	99	98	64	46
6	Ethyl 4-methylpentanoate	5[b*]	872	314	305	239	114	129	110	0
7	3-Methyl-1-butanol	179,000[a]	2	1	1	1	1	< 1	< 1	< 1
8	Ethyl hexanoate	55[a]	46,521	21,033	19,863	15,985	15,655	12,340	12,984	7653
9	Hexyl acetate	5560[g]	4	2	2	1	2	< 1	< 1	< 1
10	Cyclopentanol	4200,000[b#]	0	0	0	0	0	< 1	< 1	< 1
11	Propyl hexanoate	12,800[g]	2	< 1	1	< 1	1	< 1	< 1	0
12	Ethyl lactate	128,000[a]	9	7	9	12	12	15	13	12
13	1-Hexanol	5370[a]	31	38	43	54	53	69	51	23
14	Hexyl butanoate	5000[b#]	3	2	2	1	2	1	2	< 1
15	Ethyl octanoate	13[a]	3934	1755	2673	1509	3181	1579	2957	298
16	Acetic acid	160,000[a]	< 1	0	< 1	< 1	< 1	< 1	< 1	< 1
17	Isopentyl hexanoate	1400[a]	4	1	3	2	5	2	4	0
18	Furfural	44,000[e]	< 1	0	< 1	< 1	1	3	3	3
19	2-Acetylfuran	58,504[e+]	0	0	0	0	0	< 1	< 1	< 1
20	Ethyl nonanoate	3150[a]	< 1	0	< 1	0	0	0	0	0
21	Hexyl hexanoate	1890[d]	14	6	14	7	23	11	18	< 1
22	Butanoic acid	964[a]	68	42	67	82	107	159	165	195
23	Ethyl decanoate	1120[a]	4	1	3	2	7	< 1	8	< 1
24	Phenylacetaldehyde	262[d]	8	7	13	21	26	49	63	72
25	2-Furanmethanol	2000[e#]	3	2	3	4	5	9	8	11
26	3-Methylbutanoic acid	1050[a]	5	2	4	6	12	17	18	24
27	Diethyl butanedioate	353,000[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1
28	3-Methylthiopropanol	2110[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1
29	Pentanoic acid	389[a]	17	7	14	19	29	46	56	76
30	Ethyl phenylacetate	407[f]	12	12	19	28	52	71	82	73
31	4-Methylpentanoic acid	144[d]	0	0	0	3	3	4	5	6
32	Phenethyl acetate	909[c]	< 1	0	1	2	2	4	5	6
33	Hexanoic acid	2520[a]	45	23	41	52	70	88	74	79
34	Ethyl 3-phenylpropanoate	125[a]	22	21	33	41	96	111	163	138
35	2-Phenylethanol	28,900[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	1
36	Heptanoic acid	13,800[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1
37	2-Methoxy-*p*-cresol	50[b—]	0	0	2	4	4	6	0	0
38	Phenol	18,900[c]	< 1	< 1	< 1	< 1	0	< 1	< 1	< 1
39	4-Ethyl-2-methoxyphenol	7[e#]	113	64	90	167	255	426	502	592
40	*p*-Cresol	60[b—]	8	7	7	11	181	25	29	33

Aroma compounds: Aroma active compounds with odor threshold. [a] Odor thresholds taken from Ref. Fan et al. (2015); [b]Odor thresholds taken from Ref. Gemert (2003); [c] Odor thresholds taken from Ref. Gao et al. (2014); [d] Odor thresholds taken from Ref. Zhao et al. (2018); [e] Odor thresholds taken from Ref. Zheng et al. (2016); [f] Odor thresholds taken from Ref. Sha (2017); [g] Odor thresholds taken from Ref. Wang et al. (2014); * Odor thresholds were determined in water; # Odor thresholds were determined in air; + Odor thresholds were determined in 30% ethanol/water solution; — Odor thresholds were determined in 20% ethanol/water solution; other odor thresholds were determined in 46% ethanol/water solution.

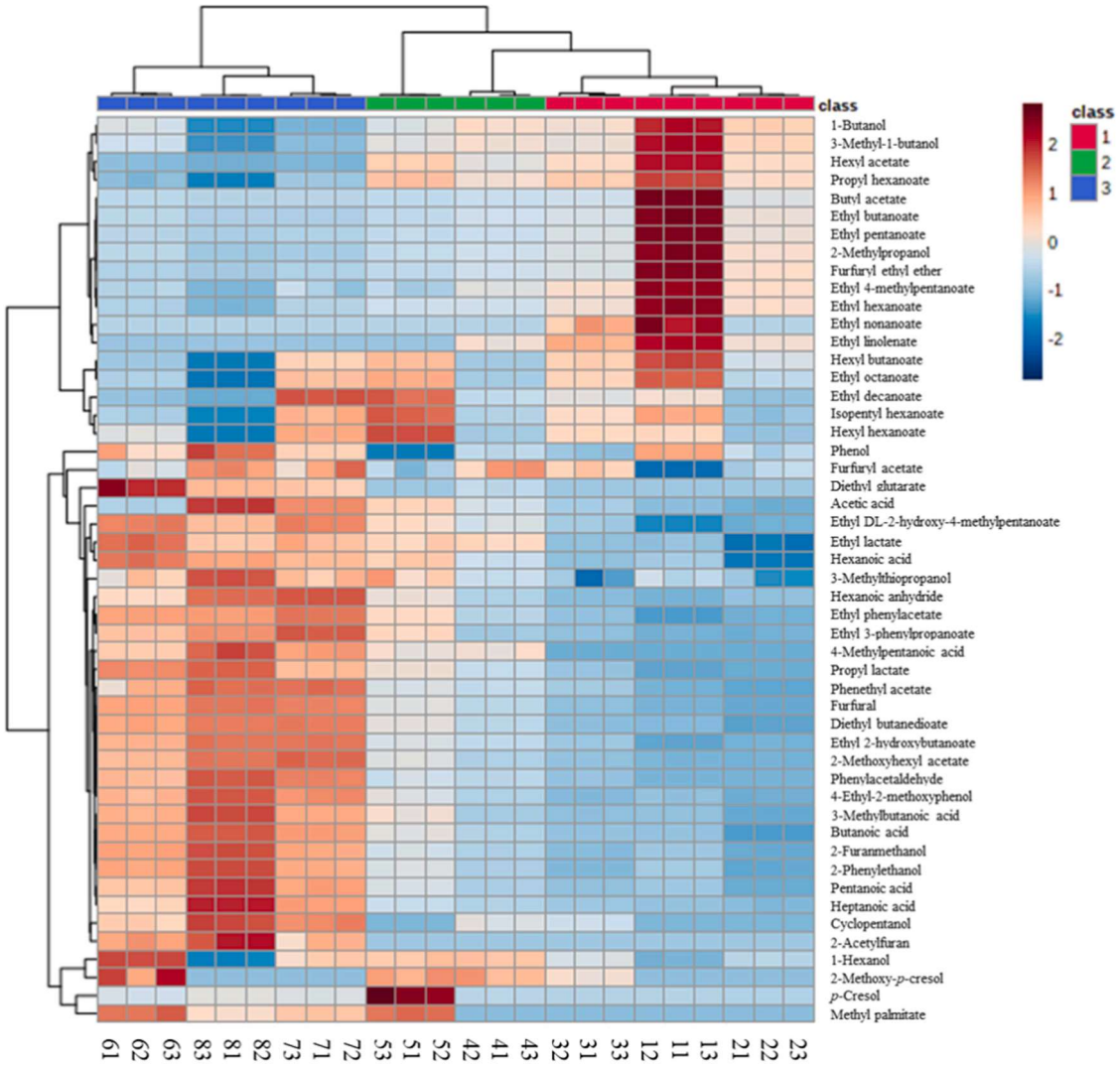


Fig. 4. Heatmap for the 50 aroma active compounds and HCA of the 8 raw Baijiu samples.

Moreover, cyclopentanol (fusel oil), propyl hexanoate (fruity, plastic), acetic acid (vinegar), 2-acetylfuran (cocoa), ethyl nonanoate (wood), diethyl butanedioate (fruity), 3-methylthiopropanol (onion, mushroom), 2-phenylethanol (rosy, honey), heptanoic acid (wood), and phenol (phenolic, plastic, rubber) had lower OAV values. The OAVs of most of these compounds were less than 1.

3.5. Characteristic aroma components of strong flavor type Baijiu in different distillation stages

To visualize the relationship between the Baijiu samples from different distillation stages and the 50 aroma compounds, a heat map and hierarchical cluster analysis (HCA) were used for the correlation, as shown in Fig. 4. By cluster analysis, the Baijiu samples in the first stage of distillation (0–6 min), in the middle stage of distillation (6–10 min), and in the post-distillation stage (10–16 min) were gathered together separately. The concentration of aroma active compounds in the Baijiu samples at different distillation stages varied greatly. 1-Butanol, 3-methyl-1-butanol, hexyl acetate, propyl hexanoate, butyl acetate, ethyl butanoate, ethyl pentanoate, 2-methylpropanol, furfuryl ethyl ether, ethyl 4-methylpentanoate, ethyl hexanoate, ethyl nonanoate, ethyl linolenate, hexyl butanoate, and ethyl octanoate were higher in the samples in the early stage of distillation and the extension of the distillation time basically showed a gradually decreasing trend. Most of these compounds were esters and alcohols with small molecular weight and volatile, and they mostly contributed a strong fruity and sweet flavor to the aroma of Baijiu; ethyl decanoate, isopentyl hexanoate, hexyl hexanoate. methyl palmitate, and *p*-cresol were higher in Baijiu samples in the middle of distillation; phenol, furfuryl acetate, diethyl glutarate, acetic acid, ethyl DL-2-hydroxy-4-methylpentanoate, ethyl lactate, hexanoic acid, 3-methylthiopropanol, hexanoic anhydride, ethyl phenylacetate, ethyl 3-phenylpropanoate, 4-methylpentanoic acid, propyl lactate, phenethyl acetate, furfural, diethyl butanedioate, ethyl 2-hydroxybutanoate, 2-methoxyhexyl acetate, phenylacetaldehyde, 4-ethyl-2-methoxyphenol, 3-methylbutanoic acid, butanoic acid, 2-furanmethanol, 2-phenylethanol, pentanoic acid, heptanoic acid, cyclopentanol, 2-acetylfuran, 1-hexanol, and 2-methoxy-*p*-cresol were mainly present in the samples in the later stage of the distillation, and they tended to increase gradually with the extension of the distillation time. These compounds were mostly esters, acids and aromatic compounds with relatively large molecular weight and difficult to volatile. Most of them brought fruity, sweaty, rancid and floral aromas to the raw Baijiu.

3.6. Exploration of key aroma active compounds

PCA was carried out based on the aroma compounds data to reveal the distribution of the samples, of which the PCA score and loading plots were shown in Fig. 5. A trend of intergroup separation was shown on the

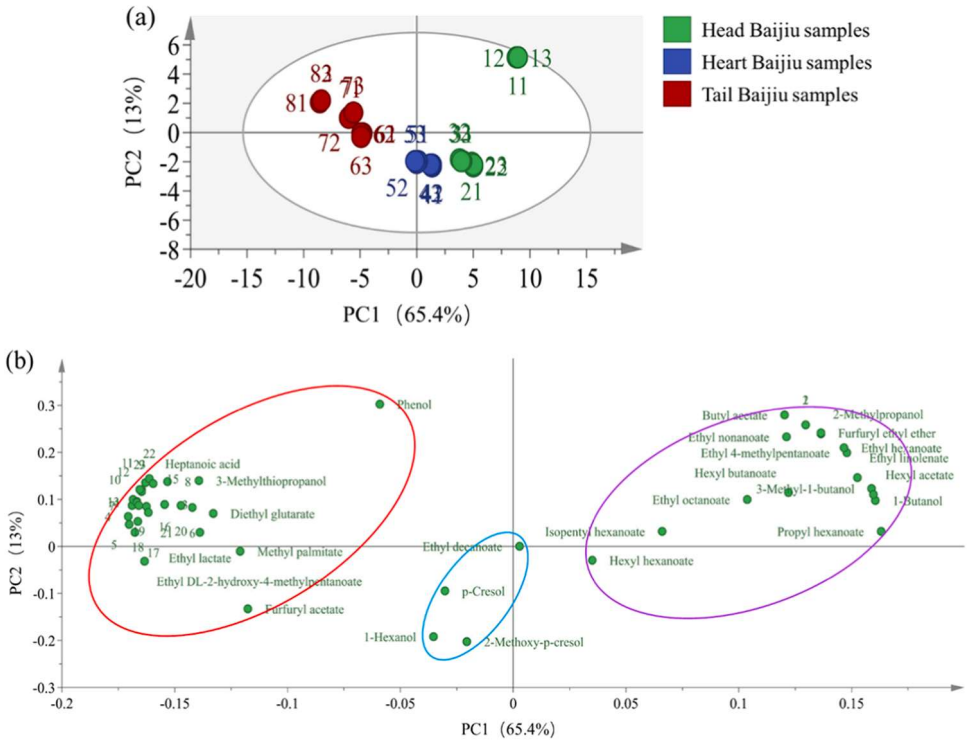


Fig. 5. PCA analysis based on aroma active compounds: (a) scores and (b) loading plot of PCA for classification of samples from different distillations. Green, blue and red dots represent the head, heart and tail Baijiu samples, respectively. (The numerical labels 1–23 in the loading plot represent the compounds ethyl butanoate, ethyl pentanoate, cyclopentanol, ethyl 2-hydroxybutanoate, propyl lactate, acetic acid, furfural, 2-acetylfuran, butanoic acid, phenylacetaldehyde, 2-furanmethanol, 3-methylbutanoic acid, diethyl butanedioate, 2-methoxyhexyl acetate, pentanoic acid, hexanoic anhydride, ethyl phenylacetate, 4-methylpentanoic acid, phenethyl acetate, hexanoic acid, ethyl 3-phenylpropanoate, 2-phenylethanol and 4-ethyl-2-methoxyphenol, respectively).

score plot of data obtained in both positive and negative modes. The principal components (PCs) PC1 and PC2 represented 65.4% and 13.0% of the total variance, respectively, with the cumulative contribution rate of the first two PCs accounting for 78.4%, which indicates that they were sufficient to explain the total variance in the dataset. The Baijiu samples clustered into three distinct groups in the dimensional graph based on the PC1 × PC2 score, as shown in Fig. 5(a). PC1 was positively correlated with the head Baijiu samples but negatively correlated with the tail Baijiu samples. These results showed that there were differences between the Baijiu samples at different distillation stages.

Additional information can also be found in the loading plot in Fig. 5 (b). As pictured, PC1 was strongly associated with the compounds in the purple circle, indicating higher concentrations in samples that showed positive scores (the head Baijiu). On the other hand, the compounds in the red circle had a negative correlation with PC1, while the tail Baijiu samples with negative scores on PC1 corresponded to high concentrations of those compounds. The compounds in the blue circles correspond to the heart Baijiu samples in the scores plot, indicating that the content of these compounds in the heart Baijiu samples was higher. Furthermore, in order to reduce the overlap of compounds in the loading plot, some compounds were represented by numbers and the compounds represented by all numbers had been noted in the figure caption.

The results obtained above showed that the different distillations of baijiu caused significant differences with respect to the aroma compound profiles.

As a supervised multivariate statistical analysis, PLS-DA can be considered a good model for biological data and provide a powerful approach to discriminate samples with different characteristics. Thus, PLS-DA models were used in this study to identify specific marker compounds that led to the differences in the aromas of 8 Baijiu samples according to the data matrices of the detected aroma active compounds at these concentrations and OAVs. A variable importance in projection (VIP) value variable selection method was applied to identify the most statistically significant variables. First, logarithmic transformation was used for all the detected compounds, giving values in a smaller range without masking the effects of small values within the data. For concentration analysis, as seen in Fig. 6(a), 89.9% of the total variance with R2Y = 99.3% and Q2 = 98.7% was explained, and three groups of samples were successfully differentiated in the quadrant diagram: (a) first and fourth quadrant sample dots for the head samples; (b) second quadrant sample dots for the heart samples; and (c) third quadrant sample dots for the tail samples. In addition, permutation tests (n = 200) were performed to evaluate whether the discriminant models were overfitting the data. The permutation tests randomly rearranged the experiments by changing the sort order of the classification variables (Y) and randomly assigned Q2Y up to 200 times. The recommended values for good fitting of models have been described as all blue Q2 values to the left are lower than the original points to the right or the blue regression line of the Q2 points intersects the vertical axis (on the left) at or below zero. Herein, the permutation test results met the above conditions, which indicated that the models were not overfitting the data. So the initial model outperformed the randomly permuted models (Fig. 6b). Then 12 different compounds were selected from the initial 50 detected aroma active compounds that showed high VIP values (VIP > 1.0) as seen in the red frame in Fig. 6(c), namely, ethyl hexanoate, 1-hexanol, acetic acid, ethyl butanoate, ethyl DL-2-hydroxy-4-methylpentanoate, 1-butanol, furfural, butanoic acid, 3-methyl-1-butanol, hexanoic acid, and diethyl butanedioate.

By contrast, as seen in Fig. 6(d), 96.7% of the total variance with R2Y = 98.8% and Q2 = 97.6% was explained by OAVs, and three groups of samples were successfully differentiated in the quadrant diagram similar to the results above. Also, the permutation test results indicated that the models were not overfitting the data and the initial model outperformed the randomly permuted models (Fig. 6(e)). Moreover, 10 marker compounds, as seen in the red frame in Fig. 6(f), namely, ethyl hexanoate, *p*-cresol, ethyl butanoate, 4-ethyl-2-methoxyphenol, ethyl octanoate, ethyl pentanoate, ethyl 4-methylpentanoate, 1-hexanol, butanoic acid, and ethyl 3-phenylpropanoate were identified with high VIP values (VIP > 1.0).

To make the results more convincing and the selected key substances more representative, the compounds common to the two VIP results were taken as the final key compounds. Overall, 4 discriminant compounds, namely, ethyl hexanoate, ethyl butanoate, 1-hexanol, and butanoic acid were identified as the most potent volatiles contributing to

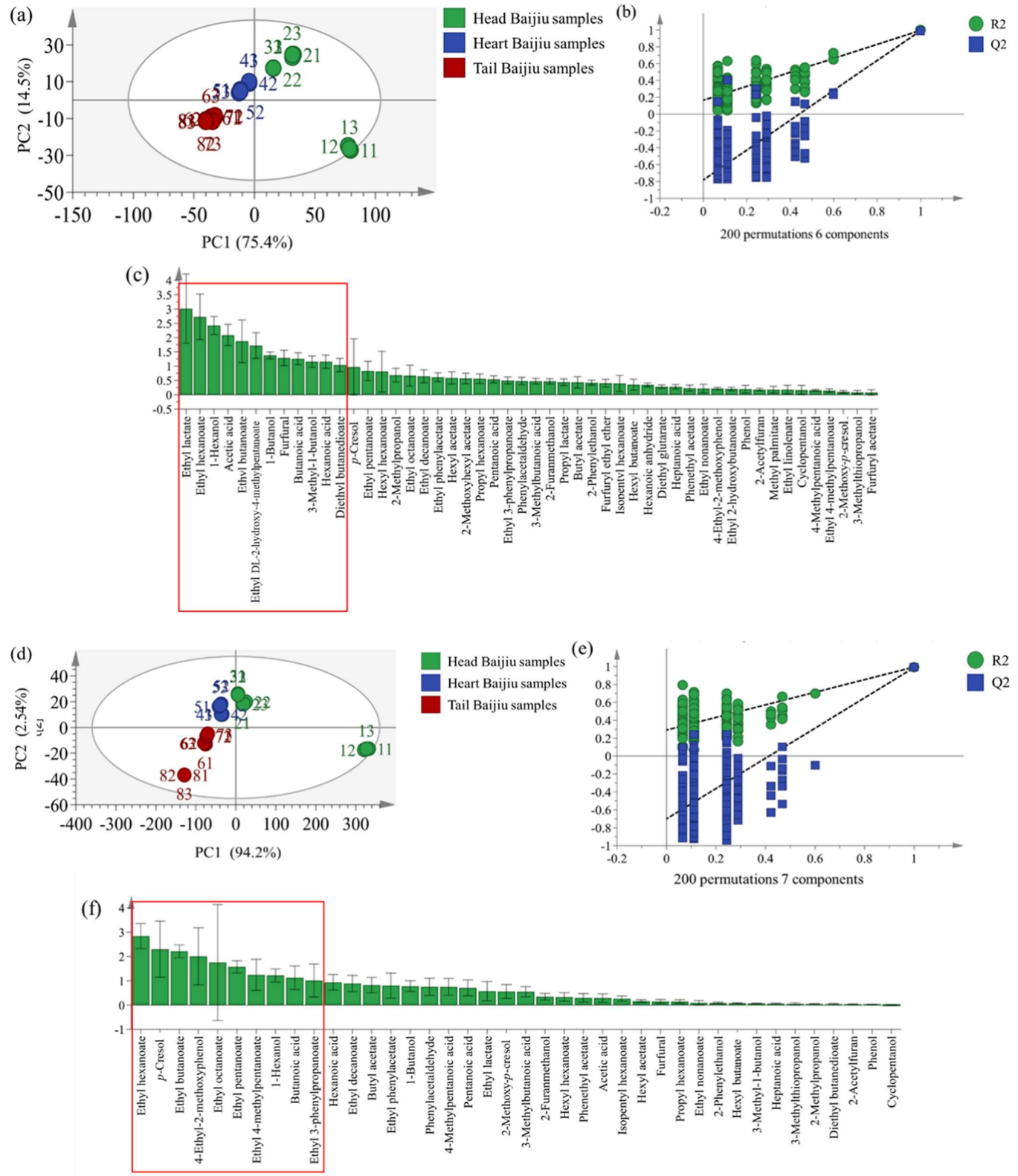
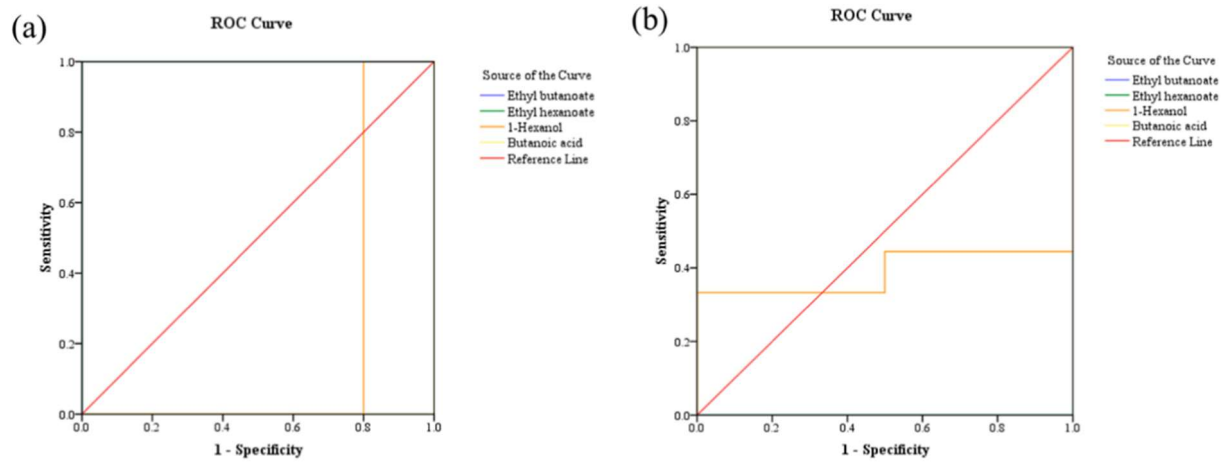


Fig. 6. PLS-DA analysis based on aroma active compounds: (a) scores, (b) permutation test and (c) VIP plot using the concentrations of aroma active compounds, which summarized the importance of the variables to both explain the aroma active compounds and correlate with Baijiu samples, sorted from high to low; (d) scores, (e) permutation test and (f) VIP plot using OAVs of aroma active compounds. Green, blue and red dots represent the head, heart and tail Baijiu samples, respectively.

discriminating the aroma perception of the raw Baijiu samples at different distillation stages.

In order to judge the diagnostic ability of the selected key aroma active compounds in distinguishing Baijiu samples at different distillation stages, the data was analyzed by ROC curve using SPSS. First, we judged the ability of these compounds to distinguish the head Baijiu samples and other Baijiu samples from heart and tail distillation stages. As shown in Fig. 7(a), the diagnostic ability of ethyl butanoate and ethyl



Area Under the Curve

Test Result Variable(s)	Area	Std. Error[a]	Asymptotic Sig.[b]	Asymptotic 95% Confidence Interval	
				Lower Bound	Upper Bound
Ethyl butanoate	1.000	.000	.000	1.000	1.000
Ethyl hexanoate	1.000	.000	.000	1.000	1.000
1-Hexanol	.200	.103	.016	-.002	.402
Butanoic acid	.000	.000	.000	.000	.000

a. Under the nonparametric assumption

b. Null hypothesis: true area = 0.5

Area Under the Curve

Test Result Variable(s)	Area	Std. Error[a]	Asymptotic Sig.[b]	Asymptotic 95% Confidence Interval	
				Lower Bound	Upper Bound
Ethyl butanoate	.000	.000	.001	.000	.000
Ethyl hexanoate	.000	.000	.001	.000	.000
1-Hexanol	.389	.156	.480	.083	.694
Butanoic acid	1.000	.000	.001	1.000	1.000

a. Under the nonparametric assumption

b. Null hypothesis: true area = 0.5

Fig. 7. ROC analysis based on 4 discriminant compounds: (a) ROC curve and area under the curve to judge the distinguishability of the head Baijiu samples and other Baijiu samples from heart and tail distillation stages, (b) ROC curve and area under the curve to judge the distinguishability of the heart Baijiu samples and the tail Baijiu samples.

hexanoate both reached 100%, while 1-hexanol and butanoic acid had almost no diagnostic ability. Therefore, it was concluded that ethyl butanoate and ethyl hexanoate could be used as different markers to distinguish the head Baijiu samples and other Baijiu samples from the heart and tail distillation stages. Secondly, we judged the diagnostic ability of these compounds in distinguishing the heart Baijiu samples and the tail Baijiu samples. As shown in Fig. 7(b), the diagnostic ability of butanoic acid reached 100%, while ethyl butanoate, ethyl hexanoate and 1-hexanol had almost no diagnostic capability. Therefore, it was concluded that butyric acid could be used as a difference marker for distinguishing the heart Baijiu samples and the tail Baijiu samples.

Through further analysis, it was found that ethyl butanoate and ethyl caproate were easy to be steamed out at the early distillation stage due to their small molecular weight, easy ethanol, and volatile properties and the content of these two compounds gradually decreased with the extension of the distillation time. The sum of the content of the two compounds in the head Baijiu samples was above 1000 mg/L, which was different from the heart Baijiu samples and tail Baijiu samples (below 1000 mg/L). Therefore, these two compounds were the most effective compounds for distinguishing the head Baijiu samples from other Baijiu samples. In addition, because of its higher boiling point, butyric acid was concentrated in the later stage of distillation and gradually increased with the extension of the distillation time. Its content in the tail Baijiu sample was more than 150 mg/L, which was different from the heart Baijiu samples and the head Baijiu samples (almost below 100 mg/L). Therefore, this compound was the most effective substance used to distinguish between the heart Baijiu samples and tail Baijiu samples.

4. Conclusion

The changing trends of the aroma compounds in raw Baijiu samples from the whole distillation process were investigated by GC-MS and GC-IMS. There were great differences between the different distillations of Baijiu samples. Esters and alcohols were more abundant in the Baijiu samples from the early distillations, but acids were mostly concentrated in the late distillations. Most alcohol-soluble and low molecular weight compounds showed a downward trend with the extension of distillation time, but water-soluble, high molecular weight and high boiling compounds showed the opposite result. Moreover, the differences in compounds identification between the two detection methods were compared, and GC-IMS was more sensitive to the response of aldehydes and ketones. Then, 50 compounds were sniffed by GC-O-MS, which were analyzed by heatmap and HCA to efficiently reveal three categories of raw Baijiu within different distillations and the aroma active compounds with different properties also cluster with each other. Finally, PCA and PLS-DA were used to find the aroma active markers related to the classification of Baijiu samples at different distillation stages, and ROCs were further used to verify that ethyl butanoate and ethyl hexanoate were the most effective difference marker to distinguish the head Baijiu samples from the Baijiu samples from other distillation stages; butanoic acid was the most effective difference marker for distinguishing the heart Baijiu samples from the tail Baijiu samples.

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CRediT authorship contribution statement

Fei He: Investigation, Methodology, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Shiqi Yang:** Methodology, Formal analysis, Data curation. **Guihu Zhang:** Formal analysis, Data curation. **Ling Xu:** Conceptualization, Samples acquisition, Supervision. **Hehe Li:** Conceptualization, Writing – review & editing. **Jinyuan Sun:** Conceptualization, Writing – review & editing. **Huangming Quan:** Conceptualization, Writing – review & editing. **Fuping Zheng:** Conceptualization, Writing – review & editing. **Baoguo Sun:** Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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